PHY101: Introduction to Physics-1

Monsoon, 2018

Tutorial-12

1. At 25 m below the surface of the sea (, ρ = 1025 kg/m^3), where the temperature is 5°C, a diver exhales an air bubble having a volume of 1 cm^3. If the surface temperature of the sea is 20°C, what is the volume of the bubble just before it breaks the surface?

2. To a first approximation, the sun consists of a gas of equal numbers of protons and electrons. The temperature at the centre of the Sun is about 1x10^7 K, and the density of the Sun is about 1x10^5 kg/m^3. Because the temperature is so high, the protons and the electron are separate particles (rather than being joined together to form hydrogen atoms). Estimate the pressure at the centre of the Sun.

3. A gas is at 0°C. If we wish to double the speed of its molecules, to what temperature must the gas be brought?

4. If a mixture of n_1 moles of monatomic gas and n_2 moles of diatomic gas has γ =1.5, then what will be the ratio of n_1 and n_2?

5. Consider 2 mol of an ideal diatomic gas. (a) Find the total heat capacity of the gas at constant volume and at constant pressure assuming the molecules rotate but do not vibrate. (b) Repeat the calculation assuming the molecules both rotate and vibrate.